

SEDAR 73 South Atlantic Red Snapper

Research recommendations

Results of this assessment are sensitive to natural mortality. The scale and age dependence of natural mortality were estimated using meta-analytical methods, as is common in SEDAR assessments. While such methods describe relationships between M and other life-history characteristics (growth, maximum age) averaged across species, they may not describe well the natural mortality of any particular species. Research into natural mortality specific to red snapper in the South Atlantic region would benefit this assessment.

This assessment used two indices of abundance from SERFS, one from chevron trap data and one from video data. In previous SEDAR assessments, indices from these gears were combined prior to fitting within the assessment, because the two gears do not sample independent (cameras housed on top of traps). However, this assessment includes these as separate indices because of new information suggesting that their selectivities differ for red snapper. To account for dependence of sampling, the likelihoods of the indices were multiplied each by 0.5, such that the two combined received full weight. Research into statistical methodology for fitting indices derived from non-independent sampling, such as with a bivariate likelihood, might benefit this and other assessments.

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Sampling Recommendations

The selectivity of video gear was inferred for this assessment, based on sampling by FWRI and by SERFS (separately) using stereo cameras. Increased sampling with stereo video cameras would benefit this and other assessments of South Atlantic reef fishes, particularly for cases where selectivity of video cameras cannot reasonably be assumed to mirror that of chevron traps.

Estimates of recreational landings and discards are critical for this assessment and for monitoring the stock. Thus, it remains important to continue estimation of recreational landings during mini-seasons and discards year-round, and any potential methodological or sampling improvements should be implemented if possible.